

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)
Master of Business Administration (MBA)
Post Graduate Diploma in Management (PGDM)
Semester –II University Examination – May 2019
MB742.3 Quantitative Analysis for Management – 2 (New)
15.05.2019, Wednesday Time: 10:00 AM to 01:00 PM Maximum Marks: 70

Instructions:

1. This paper contains six questions.
2. All questions are compulsory.
3. Each of the first five questions will carry 10 marks and the last question will carry 20.
4. Tables / Graphs shall be provided by the institute / university.

- Q-1 (a)** A pineapple firm produces two products A and B. The specific amounts of material, labour and equipment required to produce each product and the availability of each of these resources are shown in the table below: **05**

<i>Particular</i>	<i>A</i>	<i>B</i>	<i>Available Resources</i>
Labour (man – hours)	3	2.0	12.0
Equipment (Machine Hours)	1	2.3	6.9
Material (Unit)	1	1.4	4.9

Assume one unit of A and B gives a profit of Rs. 2 and Rs. 1, respectively. Formulate this as LPP and solve graphically.

- (b)** A department has five employees with five jobs to be performed. The time (in hours) each man will take to perform each job is given in the cost matrix: **05**

<i>Jobs</i>	<i>Employees</i>				
	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>
<i>A</i>	10	5	13	15	16
<i>B</i>	3	9	18	13	6
<i>C</i>	10	7	2	2	2
<i>D</i>	7	11	9	7	12
<i>E</i>	7	9	10	4	12

How should the jobs be allocated, one per employee, so as to minimize the total man – hours?

- Q-2** A product is produced by four factories F_1 , F_2 , F_3 and F_4 . Their unit production cost is Rs. 2, 3, 1 and 5 respectively. Production capacity of the factories is 50, 70, 30 and 50 units, respectively. The product is supplied to four stores S_1 , S_2 , S_3 and S_4 , the requirement of which are 25, 35, 105 and 20, respectively. Unit cost of transportation are given in the table below: **10**

<i>Factories</i>	<i>Stores</i>			
	<i>S₁</i>	<i>S₂</i>	<i>S₃</i>	<i>S₄</i>
<i>F₁</i>	2	4	6	11
<i>F₂</i>	10	8	7	5
<i>F₃</i>	13	3	9	12
<i>F₄</i>	4	6	8	3

Formulate the transportation problem and determine the optimum solution using:

1. North West Corner Method
2. Least Cost Method
3. Vogel's Approximation Method (VAM)

Also identify the best method which minimizes the transportation cost.

- Q-3** A cabinet manufacturing company is planning to introduce a new model of cabinets which requires the following tasks: **10**

Task	Description	Task Time (Minutes)
A	Prepare the wheels	10
B	Mount the wheels	5
C	Assemble the sides	15
D	Attach the top	11
E	Attach the base	10
F	Insert the brackets	5
G	Insert the shelves	5
H	Attach the doors	10
I	Attach the back panel	10
J	Paint the unit	15

The wheels are mounted after they are prepared. The base cannot be attached until the slides are assembled and the wheels are mounted. The top cannot be attached nor the brackets inserted until the sides are assembled. The selves are inserted after the brackets are installed. The back panel is attached after the base and top are attached. The doors are attached after the shelves are inserted and the top and base are attached. The unit is painted after the back and doors are attached.

You are required to:

- Identify the immediate predecessors of each task and draw the network.
- Find the critical path(s) and list the critical activities.
- Obtain the earliest and latest start and completion times of all the activities and their total floats.

- Q-4** A tourist information desk has a single server, who answers questions for tourists, who stand in line to be served. Assume the time it takes to answer a tourist's question(s) is exponentially distributed with mean of 3 minutes. Tourists arrive to the desk at a rate of 18 per hour. **10**
Assuming that the pattern of tourists arrival is Poisson distributed and the servicing time is exponentially distributed, determine the following:

- The utilization parameter
- The probability that the queuing system is idle
- Expected number of tourists waiting at information desk
- Expected number of tourists waiting in a queue
- Expected time spent by a tourists at information desk
- Expected time spent by a tourists in a queue

- Q-5** Mr. Anish had to decide whether or not to drill on his farm. In his own town, only 40% of the wells drilled were successful at 200 feet of depth. Some of the farmers who did not get water at 200 feet, drilled further upto 250 feet but only 20% struck water at 250 feet. Cost of drilling is Rs. 50 per foot. **10**

Mr. Anish estimated that he would pay Rs. 18000 during 4 – year period in the present value terms, if he continues to buy water from the neighborhood rather than go for the well which would have a life of 4 years. Mr. Anish has three decisions to make:

- Should he drill upto 200 feet?
- If no water is found at 200 feet, should he drill upto 250 feet?
- Should he continue to buy water from his neighbor?

Draw the decision tree and determine the best strategy.

Q-6

20

A dietician wants to simulate arrival of her patients and her consultation time with the following random numbers. Her assistant has already prepared the random number allocation tables. The dietician wants to have an idea of her ideal time and patients' waiting time. She starts her consultation at 10:00 am and wants to give an appointment at an interval of 20 minutes.

The Random Number table is as follows:

Arrival of Patients	15	04	35	67	75	66	25
Consultation Time (minutes)	17	15	12	58	60	72	30

Random Number Allocation Table

Patient Punctuality	Probability	Consultation Time	Probability
Minutes Early 3	0.05	15	0.13
2	0.18	18	0.15
1	0.40	20	0.28
On time	0.25	25	0.34
Minutes Late 2	0.08	30	0.10
4	0.04		

You are required to simulate:

- Simulate the arrival and consultation times and find out the dietician's idle times and patients' waiting times.
- If clients are sensitive to waiting, how would you advise the dietician as a Management Consultant, based on the results of your exercise?